

The $a = 3$ slice of two permutational Ramsey conjectures:

$$R_{\text{dih}}(P_3^{\text{alt}}, K_b) = R_{\text{cyc}}(P_3^{\text{alt}}, K_b) = 2b - 1 \text{ for all } b$$

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Draft v1 — 2026-08-12

Abstract

For $b \in \mathbb{N}$ let $R_{\text{dih}}(P_3^{\text{alt}}, K_b)$ denote the dihedral Ramsey number, in the sense of Damnjanović–Đorđević [1], of the alternating path of order 3 against the complete graph K_b . We prove $R_{\text{dih}}(P_3^{\text{alt}}, K_b) = 2b - 1$ for all b — the $a = 3$ slice of [1, Conjecture 4.9] — and $R_{\text{cyc}}(P_3^{\text{alt}}, K_b) = 2b - 1$ for all b — the $a = 3$ slice of [2, Conjecture 4.23]. The dihedral case rests on an order argument special to three vertices: $\text{Dih}(3) = \text{Sym}(3)$, so rotations and reflections already exhaust all relabelings, dihedral embeddability collapses to ordinary subgraph containment, and Chvátal’s 1977 tree-complete-graph theorem applies. The cyclic case does not follow from the reflection-symmetry reduction of [1, Corollary 2.5] — P_3^{alt} lacks reflection symmetry — but does follow from an orbit-equality argument via [1, Proposition 2.4]. A block-partition construction, valid for every connected pattern and every permutation group, gives the matching lower bound.

The identity was corroborated by direct SAT computation on the raw dihedral encoding for $b = 2, \dots, 7$, and $b = 3$ is fully machine-verified by the accompanying standalone script; the lower bound for every b , and the $\text{Dih}(3) = \text{Sym}(3)$ collapse, are additionally machine-verified in Lean 4 (the upper bound is the citation). The upper bound for patterns of order $a \geq 4$ remains open.

1 History and the two conjectures

Damnjanović and Đorđević [1] compute small *reflective* and *dihedral* Ramsey numbers — permutational Ramsey numbers under reflection or dihedral symmetry groups — by SAT encoding under fixed solver budgets, and state several closed-form conjectures that their own tables support only as computed cells or unproved lower bounds. The one this note closes at $a = 3$ reads, verbatim:

Conjecture 4.9 [1]. For any $a, b \in \mathbb{N}$, we have $R_{\text{dih}}(P_a^{\text{alt}}, K_b) = 1 + (a - 1)(b - 1)$.

For the *cyclic* variant of the same pairing, [1]’s Table 2 records the matching conjecture from the companion paper of Bašić, Damnjanović, Stevanović, and Stošić, citing it as [2, Conjecture 4.23]:

Conjecture 4.23 [2]. $R_{\text{cyc}}(P_a^{\text{alt}}, K_b) = 1 + (a - 1)(b - 1)$ for $a, b \in \mathbb{N}$.

At $a = 3$ both right-hand sides equal $2b - 1$.

What is and is not new here. [2]’s companion repository already tabulates cyclic values matching $2b - 1$ for $b = 3, \dots, 8$ as ordinary computed SAT cells (we traced them to the underlying solver logs, not just the summary table), without stating a closed form. The new content of this note is therefore: the closed form itself (all b , not six data points), its proof (a citation chain plus two group-theoretic facts, not a search), and — for the *dihedral* value, which [2] never tabulates directly — a new all- b closed form. The $b = 3, \dots, 8$ cyclic arithmetic is not new.

2 Definitions

All definitions are restated from [1, §§1–2].

Permutational Ramsey numbers. For graphs $H_1, \dots, H_k$ on vertex sets $\{0, \dots, |H_j| - 1\}$ and permutation groups Γ_j on $V(H_j)$, the graph H_j is Γ_j -**embeddable** in a graph G (on a totally ordered vertex set) if there is a homomorphism $H_j \rightarrow G$ of the form $\psi \circ \varphi$, where $\varphi \in \Gamma_j$ and $\psi : V(H_j) \rightarrow V(G)$ is an **increasing** injection. Only the pattern's own vertices may be permuted (by φ); the placement ψ into the host's fixed vertex order must be order-preserving. The **permutational Ramsey number** $R(H_1^{\Gamma_1}, \dots, H_k^{\Gamma_k})$ is the least n such that every k -edge-coloring of K_n contains, for some color j , a Γ_j -embedded copy of H_j in color j .

The increasing-injection requirement is the substance of the definition: arbitrary relabelings of the host K_n are *not* a symmetry of the problem — only the pattern's own group Γ_j is. This is exactly why the choice of group matters and why the theorem below is not vacuous.

The groups. For a pattern of order m on $\{0, \dots, m - 1\}$, the **dihedral group** $\text{Dih}(m) = \langle \sigma, \rho \rangle$ where $\sigma(i) = i + 1 \pmod{m}$ is the rotation and $\rho(i) = m - 1 - i$ the reflection; $|\text{Dih}(m)| = 2m$. The **cyclic group** $\text{Cyc}(m) = \langle \sigma \rangle$ has order m . R_{dih} (resp. R_{cyc}) denotes the permutational Ramsey number in which every Γ_j is the dihedral (resp. cyclic) group of its own pattern H_j . Taking every Γ_j to be $\text{Sym}(|H_j|)$ recovers the classical Ramsey number [1, §1]: with all relabelings available, any ordinary subgraph copy can be pre-permuted so that its placement is increasing.

Γ -isomorphism. $F \cong_{\Gamma} H$ iff some isomorphism from F to H is an element of Γ [1, §2]; its equivalence classes are the Γ -orbits of labeled patterns. Two facts from [1] are used below:

Proposition 2.1 [1]. Let $H_1, \dots, H_k$ be graphs and $\Gamma_1, \dots, \Gamma_k$ permutation groups on their respective vertex sets. If $F_j \cong_{\Gamma_j} H_j$ for each j , then $R(F_1^{\Gamma_1}, \dots, F_k^{\Gamma_k}) = R(H_1^{\Gamma_1}, \dots, H_k^{\Gamma_k})$.

Proposition 2.4 [1]. Let $H_1, \dots, H_k$ be graphs, and $\Gamma_1, \dots, \Gamma_k$ and $\Lambda_1, \dots, \Lambda_k$ permutation groups on their respective vertex sets. If, for each j , H_j lies in the same equivalence class under both $\cong_{\Gamma_j}$ and $\cong_{\Lambda_j}$, then $R(H_1^{\Lambda_1}, \dots, H_k^{\Lambda_k}) = R(H_1^{\Gamma_1}, \dots, H_k^{\Gamma_k})$.

The pattern. The alternating path P_a^{alt} is the path on $\{0, \dots, a - 1\}$ whose vertex sequence is $(0, a - 1, 1, a - 2, \dots)$, with edges between consecutive terms. For $a = 3$ the sequence is $(0, 2, 1)$, so

$$P_3^{\text{alt}} = \text{the path } 0\text{--}2\text{--}1, \quad \text{edges } \{0, 2\} \text{ and } \{1, 2\}, \quad \text{center vertex } 2.$$

3 $\text{Dih}(3)$, concretely

$\text{Dih}(3)$ is generated by $\sigma = (0 \mapsto 1 \mapsto 2 \mapsto 0)$ and $\rho = (0 \leftrightarrow 2, \text{fixing } 1)$. Its six elements, as permutations (images of $0, 1, 2$ in order):

element	$0 \mapsto$	$1 \mapsto$	$2 \mapsto$
id	0	1	2
σ	1	2	0
σ^2	2	0	1
ρ	2	1	0
$\sigma\rho$	0	2	1
$\sigma^2\rho$	1	0	2

These are six distinct permutations of a three-element set, and $\text{Sym}(3)$ has $3! = 6$ elements in total. That observation is Lemma 3 below; the point of listing the elements is that nothing is hidden — for three vertices, rotations and reflections already produce every relabeling.

4 Lemma 1: K_b -embeddability is group-independent

Lemma 1. *For any $b \in \mathbb{N}$, $n \geq b$, any graph G on $\{0, \dots, n-1\}$, and any permutation group Γ on $\{0, \dots, b-1\}$: K_b is Γ -embeddable in G if and only if G contains a clique of size b .*

Proof. ($\Leftarrow$) Let $v_0 < v_1 < \dots < v_{b-1}$ be the vertices of a b -clique of G . Take $\psi(i) = v_i$ and $\varphi = \text{id} \in \Gamma$. Every pair of K_b 's vertices is an edge, and every image pair is a clique edge.

($\Rightarrow$) $\psi \circ \varphi$ is injective (φ is a bijection, ψ an injection), and every pair of K_b 's vertices is an edge, so the b image vertices are pairwise adjacent in G — a clique. $\square$

5 Lemma 2: the block-partition lower bound

Lemma 2. *Let $a, b \geq 1$ and $n = (a-1)(b-1)$. Fix any partition of $\{0, \dots, n-1\}$ into $b-1$ blocks of size $a-1$ (contiguous or not), any connected graph H of order a , and any permutation groups Γ (on $V(H)$) and Λ (on $V(K_b)$). Color every same-block pair with color 1 and every cross-block pair with color 2. Then color 1 contains no Γ -embedded H and color 2 contains no Λ -embedded K_b ; consequently*

$$R(H^\Gamma, K_b^\Lambda) \geq 1 + (a-1)(b-1).$$

Proof. Color 1's graph is a disjoint union of $b-1$ cliques K_{a-1} . The image of H under the injective homomorphism $\psi \circ \varphi$ is a connected subgraph on a vertices, hence lies inside a single component — but each component has only $a-1 < a$ vertices. So color 1 has no Γ -embedded H .

Color 2's graph is the complete $(b-1)$ -partite complement; by Lemma 1 a Λ -embedded K_b requires a color-2 clique of size b , but any b vertices distributed among $b-1$ blocks share a block by pigeonhole, and that pair has color 1. $\square$

Specializing $H = P_a^{\text{alt}}$ (connected, order a) and $\Gamma = \text{Dih}(a)$, $\Lambda = \text{Dih}(b)$: $R_{\text{dih}}(P_a^{\text{alt}}, K_b) \geq 1 + (a-1)(b-1)$ for all $a, b \in \mathbb{N}$ — the lower-bound half of [1, Conjecture 4.9] in full generality. For $a = 3$ the witness is $b-1$ disjoint “dominoes” (red blocks of size 2) on $2b-2$ vertices; the accompanying script constructs and checks it exhaustively for $b = 2, \dots, 8$.

6 Lemma 3: $\text{Dih}(3) = \text{Sym}(3)$

Lemma 3. *As sets of permutations of $\{0, 1, 2\}$, $\text{Dih}(3) = \text{Sym}(3)$.*

Proof. $\text{Dih}(3)$'s generators are permutations of $\{0, 1, 2\}$, so $\text{Dih}(3) \leq \text{Sym}(3)$. $|\text{Dih}(3)| = 2 \cdot 3 = 6 = 3! = |\text{Sym}(3)|$, and a subgroup with the same order as its ambient finite group equals it. $\square$

Equivalently: a 3-cycle and a transposition already generate all of $\text{Sym}(3)$. This collapse is unique to $a = 3$: for every $a \geq 4$, $|\text{Dih}(a)| = 2a < a!$, and indeed $\text{Dih}(a) \neq \text{Sym}(a)$.

7 The theorem

Theorem 4. $R_{\text{dih}}(P_3^{\text{alt}}, K_b) = 2b - 1$ for all $b \in \mathbb{N}$.

Proof. By Lemma 3, $(P_3^{\text{alt}})^{\text{Dih}(3)}$ -embeddability is $(P_3^{\text{alt}})^{\text{Sym}(3)}$ -embeddability — the same group, hence literally the same relation. By Lemma 1, the group attached to the K_b argument is irrelevant, so $\text{Dih}(b)$ may be replaced by $\text{Sym}(b)$ without changing any embeddability predicate. With every group now the full symmetric group, the permutational Ramsey number is the classical one [1, §1]:

$$R_{\text{dih}}(P_3^{\text{alt}}, K_b) = R((P_3^{\text{alt}})^{\text{Sym}(3)}, K_b^{\text{Sym}(b)}) = R(P_3, K_b),$$

where P_3 denotes the (unlabeled) path of order 3 — as an unlabeled graph, P_3^{alt} is the unique tree on 3 vertices, and Proposition 2.1 (with symmetric groups, whose orbits are exactly unlabeled-isomorphism classes) makes the choice of labeled representative irrelevant. Chvátal’s tree-complete-graph theorem, restated verbatim in [1]:

Theorem 4.13 [1, quoting [3]]. For any $a, b \in \mathbb{N}$ and tree T of order a , $R(T, K_b) = 1 + (a - 1)(b - 1)$.

With $a = 3$: $R(P_3, K_b) = 1 + 2(b - 1) = 2b - 1$. □

Remark. The *lower* bound is also proved directly and independently by Lemma 2, so the only step of this note that rests on a citation rather than a self-contained argument is the upper bound $R(P_3, K_b) \leq 2b - 1$ inside Theorem 4.13.

8 The cyclic corollary — and why it is not the reflection-symmetry route

A natural first guess is [1, Corollary 2.5]: for graphs with *reflection symmetry* (the map $\varphi(x) = |H| - 1 - x$ is an automorphism), $R_{\text{dih}} = R_{\text{cyc}}$. That corollary **does not apply** to P_3^{alt} : under $\varphi(x) = 2 - x$, the edge $\{0, 2\}$ is fixed, but $\{1, 2\} \mapsto \{1, 0\} = \{0, 1\}$, which is not an edge of P_3^{alt} . P_3^{alt} lacks reflection symmetry.

The mechanism that does work is Proposition 2.4, an orbit-equality condition strictly weaker than an automorphism requirement.

Orbit computation. The $\text{Cyc}(3)$ -orbit of P_3^{alt} : applying σ to the edge set $\{\{0, 2\}, \{1, 2\}\}$ gives $\{\{1, 0\}, \{2, 0\}\}$ (center 0); applying σ^2 gives $\{\{2, 1\}, \{0, 1\}\}$ (center 1). So the orbit is the three labeled 3-paths with center 2, 0, 1 respectively. The $\text{Sym}(3)$ -orbit of P_3^{alt} is the set of *all* labeled copies of the 3-path on $\{0, 1, 2\}$ — and a labeled 3-path is determined by its center, so that orbit is the same three graphs. The two orbits are equal: a coincidence of a 3-vertex tree having only 3 labeled shapes at all, each reachable by rotation alone — nothing to do with reflection. On the other side, the orbit of K_b under any permutation group is the singleton $\{K_b\}$.

Corollary 5. $R_{\text{cyc}}(P_3^{\text{alt}}, K_b) = 2b - 1$ for all $b \in \mathbb{N}$ — the $a = 3$ slice of [2, Conjecture 4.23].

Proof. Proposition 2.4, applied coordinate-wise with $(\Gamma_1, \Lambda_1) = (\text{Cyc}(3), \text{Sym}(3))$ on P_3^{alt} (orbits equal, as computed) and $(\Gamma_2, \Lambda_2) = (\text{Cyc}(b), \text{Sym}(b))$ on K_b (orbits trivially equal), gives $R_{\text{cyc}}(P_3^{\text{alt}}, K_b) = R((P_3^{\text{alt}})^{\text{Sym}(3)}, K_b^{\text{Sym}(b)}) = R(P_3, K_b)$ — the same classical anchor as in Theorem 4, hence the same value $2b - 1$. □

The cyclic and dihedral values are thus equal, but by two different mechanisms meeting at a shared classical anchor — not by one being a one-line consequence of the other.

9 Numeric corroboration

The following computations corroborate the theorem; they are **corroboration, not proof** (the proof is §§4–8).

Direct SAT recomputation on the raw dihedral encoding (i.e., encoding $\text{Dih}(3)$ and $\text{Dih}(b)$ as-is, *not* using the Sym-collapse — an independent check of the conclusion, not just the proof steps), carried out during adversarial review:

b	predicted $2b-1$	SAT at $2b-2$	UNSAT at $2b-1$ (DRAT)	match
2	3	verified	drat-trim VERIFIED	YES
3	5	verified	drat-trim VERIFIED	YES
4	7	verified	drat-trim VERIFIED	YES
5	9	verified	drat-trim VERIFIED	YES
6	11	verified	drat-trim VERIFIED	YES
7	13	verified	drat-trim VERIFIED	YES

6/6 exact matches. A bonus check swapped K_5 ’s group from $\text{Dih}(5)$ (order 10) to $\text{Sym}(5)$ (order 120) inside the actual encoding and reproduced an identical SAT@8/UNSAT@9 boundary — a direct in-chain exercise of Lemma 1.

Accompanying script. `verify_theorem_b.py` is a standalone stdlib-Python checker that (i) builds $\text{Dih}(3)$ by generator closure — it does *not* assume Lemma 3 — and re-derives $|\text{Dih}(3)| = 6 = |\text{Sym}(3)|$; (ii) constructs Lemma 2’s witness coloring and exhaustively verifies $R_{\text{dih}}(P_3^{\text{alt}}, K_b) \geq 2b-1$ for $b = 2, \dots, 8$; and (iii) at $b = 3$ exhaustively checks all 1,024 colorings of K_5 , so $R_{\text{dih}}(P_3^{\text{alt}}, K_3) = 5$ is machine-verified in both directions by the script alone. It also checks Corollary 5’s orbit-equality premise explicitly: the $\text{Cyc}(3)$ -orbit of P_3^{alt} equals the $\text{Sym}(3)$ -orbit — the three center-labelings — and K_b ’s orbit is a singleton ($\text{Dih}(b)$ for $b = 2, \dots, 8$, all of $\text{Sym}(b)$ for $b = 2, \dots, 5$); the Proposition 2.4 application itself remains a citation.

Accompanying Lean proof. `TheoremB.lean` is a standalone Lean 4 file — core only, no mathlib, no imports — whose main theorem, `theoremB_lowerBound`, machine-checks the lower-bound half of Theorem 4 *for every $b \geq 1$ at once*: a genuine $\forall b$ theorem, not per-instance enumeration. It also proves Lemma 3 ($\text{Dih}(3) = \text{Sym}(3)$, both inclusions) and a group-invariance lemma for K_b . Axiom audit: `propext`, `Classical.choice`, `Quot.sound` only; no `native_decide`, no `sorry`.

10 What remains open

The general upper bound ($a \geq 4$) resists the two standard strategies — a diagnosis, not a hardness proof. (1) Mirroring [1]’s own pigeonhole-induction (their Theorem 4.1 mechanism): growing K_b against a fixed dihedral pattern only shows *some* embedding of P_a^{alt} exists, with no control over *which* of $\text{Dih}(a)$ ’s admissible shapes it is. (2) The classical Chvátal degree-split induction: its greedy tree-embedding lemma has the identical defect.

The $\text{Dih}(a)$ -orbit of P_a^{alt} has exactly a distinct “zigzag” shapes out of $a!/2$ labeled paths; both mechanisms certify “some shape, unrestricted,” and neither supplies leverage on which. For $a \geq 4$ the Lemma 3 collapse is unavailable ($|\text{Dih}(a)| = 2a < a!$), so the $a = 3$ route does not extend. No counterexample to Conjecture 4.9 was found; the obstruction is proof technique, not evidence against the conjecture.

11 Provenance, verification status, and scope

This work was carried out autonomously by an AI system (Claude, Anthropic) on 2026-08-11/12, as one lane of a larger Ramsey-theory research run on a single laptop. Human direction was limited to initiation and operational supervision.

Verification status, stated precisely:

- **Adversarial review** was performed by AI saboteur agents within the same pipeline — independent re-derivation of all three lemmas; direct SAT recomputation of $b = 2, \dots, 7$ on the raw dihedral encoding with DRAT certificates checked by drat-trim; a verbatim citation audit of every [1] statement used; and, for Corollary 5, an independent orbit recomputation plus exhaustive both-boundary brute force at $b = 2, 3, 4$ bypassing both Proposition 2.4 and the theorem. Zero refutations; one immaterial tally slip in motivational text was found and corrected. **No human mathematician has reviewed these claims.**
- **Partially formally verified (Lean 4).** The lower bound $R_{\text{dih}}(P_3^{\text{alt}}, K_b) \geq 2b-1$ for every $b \geq 1$ and Lemma 3 ($\text{Dih}(3) = \text{Sym}(3)$) are machine-checked in Lean 4 core (no mathlib; axioms `propext`, `Classical.choice`, `Quot.sound`; no `native_decide`), in the package’s standalone `TheoremB.lean`, which also contains the run’s Lean-proved parallel of Lemma 1 and builds on its definitions. **The upper bound is not formalized** — neither Chvátal’s theorem nor the reduction steps that reach it.
- **Chvátal 1977 is cited second-hand**, via its verbatim restatement as [1, Theorem 4.13]; the 1977 original was not consulted.
- **A disclosed reproducibility gap:** the producing lane’s original scratch scripts were never committed. Every numeric outcome the adversarial reviewer re-ran reproduced exactly, and the accompanying script re-verifies the lower bounds and the $b = 3$ cell from scratch.
- **Novelty:** same-day arXiv version checks and targeted searches (2026-08-12) found no prior statement or proof of the closed form or of the $\text{Dih}(3) = \text{Sym}(3)$ mechanism applied to this pairing. The final Semantic Scholar pass was rate-limited and relied on three same-day zero-citation fetches earlier that day.

Scope. This note treats the dihedral and cyclic cases only. The ordered and reflective variants are not part of this result. The full conjectures ($a \geq 4$) remain open (see §10).

===== NOT DONE =====

- Ordered and reflective variants of the same $a=3$ pairing: not included; this note makes no claim, in either direction, about their status.
- $a \geq 4$: nothing proved; Conjecture 4.9 remains open there (Section 10 records exactly where the standard techniques stop).
- Upper bound not formalized: the Lean file verifies the lower bound (every b), Lemma 3, and a Lemma-1 parallel, and nothing else; the upper bound rests on the citation chain through [DD26, Theorem 4.13].
- No human peer review; adversarial review was AI-internal.

WHAT IT WOULD CHANGE: a gap in the upper bound’s citation chain (the `Sym-collapse-to-classical` step, Proposition 2.1/2.4’s application, or the Chvatal statement) would invalidate the UPPER bounds beyond the SAT-verified range ($b = 2..7$ dihedral, $b = 3..8$ cyclic); the LOWER

bounds stand on kernel-checked proof for every b regardless. A prior publication of either closed form would change this note's contribution from new-result to rediscovery.

References

- [1] I. Damnjanović and I. Đorđević, *Computation of small reflective and dihedral Ramsey numbers*, arXiv:2607.06817v2 [math.CO], 12 Jul 2026.
- [2] N. Bašić, I. Damnjanović, D. Stevanović, and I. Stošić, *Some results on small ordered and cyclic Ramsey numbers*, arXiv:2604.16188v1 [math.CO], 17 Apr 2026.
- [3] V. Chvátal, Tree-complete graph Ramsey numbers, *J. Graph Theory* **1** (1977), 93. Cited via its verbatim restatement as [1, Theorem 4.13].